

Stark Shifts in Simulated Excited State Vibrational Spectra

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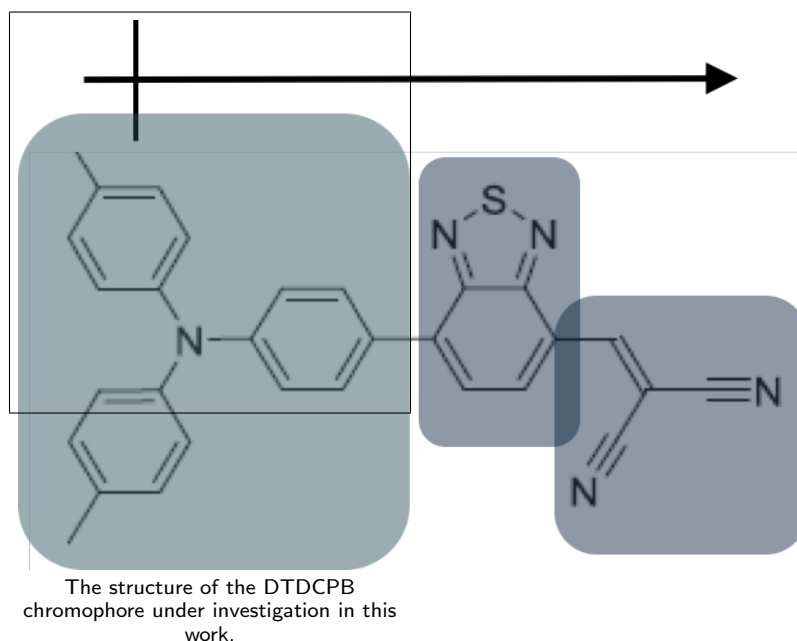
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Abstract

We have modeled the effect of external electric fields on nitrile vibrational signatures of 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3- benzothiadiazol-4-yl)methylene]propanedinitrile (DTDCPB). DTDCPB is a promising push-pull chromophore prevalent in organic photovoltaic research. By applying static electric fields *in silico* and simulating ground state IR spectra, we determined the Stark tuning rate of DTDCPB nitrile stretching to be FILLINTHEBLANK. Born-Oppenheimer molecular dynamics was used to derive the model IR spectra to correlate to experimental data. In hopes of modeling pump-probe transient IR studies of DTDCPB, computational studies of excited-state dynamics are underway. These studies will enable *in operando* monitoring of electric fields of organic photovoltaic devices.

Graphical TOC Entry



Introduction

As research in solar energy becomes increasingly important, several new strategies are being investigated to increase the efficiency of these devices. In organic photovoltaics (OPVs), one new approach uses small molecules with large ground-state dipole moments as electron donors. It is believed the inherent dipole allows for enhanced hole hopping and can minimize charge recombination (ref Griffith). One such molecule is 2-[(7-4-[N,N-Bis(4-methylphenyl)amino]phenyl-2,1,3-benzothiadiazol-4-yl)methylene] propanedinitrile (DTDCPB) and has resulted in OPVs with power conversion efficiencies of up to 9.6%. The nitrile groups present in the molecule can act as reporter moieties for vibrational spectroscopy. By modeling the effect of an applied electric field on absorption spectrum (Stark Effect), it is possible to probe the local electric field during charge transfer events in a photovoltaic device via excited state IR spectroscopy. This first requires a thorough understanding of the electric field dependence of the ground and excited state IR spectra.

Understanding the mechanisms of charge transfer events enables rational design and enhancement of current materials for novel, high-efficiency semiconductor and photovoltaic devices. Computational modeling can be a powerful tool for probing these complicated mechanisms. This study was carried out in several stages. First, the equilibrium properties of the DTDCPB molecule for the ground state were obtained through frequency calculations in *Gaussian 09*. The Stark effect was modeled by applying external static electric fields of various magnitudes and performing frequency calculations with and without the applied field. Spectral shifts were obtained from these frequency calculations and can be correlated to the presence of an electric field. Subsequently, Born-Oppenheimer Molecular Dynamics were used to obtain model spectra using a different set of methods from the standard frequency calculations in *Gaussian 09*.

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Theory

Standard Frequency Calculations

The *normal modes of vibration* are calculated by diagonalizing a matrix of the second derivatives of the electronic potential energy surface (PES) for the system of N atoms. The eigenvalues of this *Hessian* matrix correspond to the frequencies of the normal modes of vibration for the system. This matrix is obtained from the Taylor expansion about the potential energy minimum with respect to nuclear position q_k for the gradient g_k , given by Equation 1.

$$V(g_k) = V(0) + \sum_k \left(\frac{\partial V}{\partial q_k} \right) q_k + \frac{1}{2} \sum_{j,k} q_j \left(\frac{\partial^2 V}{\partial q_j \partial q_k} \right) q_k + \frac{1}{6} \sum_{j,k} q_j \left(\frac{\partial^3 V}{\partial q_j \partial q_k \partial q_l} \right) q_k q_l + \dots \quad (1)$$

where $j, k = 1, \dots, N$. We find that the Hessian is our second geometric derivative term $H_{j,k} = (\partial^2 E / \partial q_j \partial q_k)$ in the Taylor expansion, and we can more compactly rewrite Equation 1 as

$$V(g_k) = V(0) + \sum_k g_k + \frac{1}{2} \sum_{j,k} q_j H_{j,k} q_k + \frac{1}{6} \sum_{j,k} q_j A_{j,k,l} q_k q_l + \dots \quad (2)$$

Where the anharmonicity tensor element A_{jkl} is defined by the third derivative in Equation 2. Diagonalizing the Hessian yields the orthogonal set of N frequency normal modes $\{\omega_k\}$. To obtain the intensities of the transitions the vibrational intensities are given by (Equation 3). The dipole moment μ is obtained from the instantaneous electron density around the nuclei.

$$\mathcal{A}_{\tilde{\nu}}(q_k) \propto \left(\frac{\partial \mu}{\partial q_k^2} \right) \quad (3)$$

Scaling the frequencies $\{\omega_k\}$ by the intensity $\mathcal{A}_{\tilde{\nu}}(q_k)$ a simulated vibrational spectrum for the molecule.

The Stark Effect

The Stark effect is a phenomenon in which the energy level splittings are increased in the presence of an applied electric field. The effect is measured via changes to the vibrational and optical spectra of a species (**Figure?**). We can also model it computationally by performing calculations accounting for the presence of an external field. The effect is a change in dipole moment ($\Delta\mu$) dependent on the degree of charge separation for a given transition. Additionally, a change in polarizability ($\Delta\alpha$) is observed corresponding to the change in sensitivity of the vibrational transition to the applied field. The change in frequencies is modeled in Equation 4 where $V(r_k)$ is the potential with respect to position r_k of electron k .

$$\Delta\omega_k = -\Delta\vec{\mu} \cdot \left(\frac{\partial V(\vec{r})}{\partial \vec{r}_k} \right) - \frac{1}{2} \Delta\alpha \cdot \left| \frac{\partial V(\vec{r})}{\partial \vec{r}_k} \right|^2 \quad (4)$$

Born-Oppenheimer MD (BOMD)

We apply Born-Oppenheimer Molecular Dynamics (BOMD) to derive the model spectra for DTDCPB with and without the static electric field present. BOMD utilizes Born-Oppenheimer approximate time-*independent* density functional theory (DFT) calculations to compute the electronic potential at each nuclear evolution time step with classical equations of motion (EOMs) integrated by the Velocity Verlet integration method (Equations 5)

approximating the nuclear dynamics.

$$\begin{aligned}\vec{x}(t + \Delta t) &= \vec{x}(t) + \vec{v}(t)\Delta t + \frac{1}{2}\vec{a}(t)\Delta t^2 \\ \vec{v}(t + \Delta t) &= \vec{v}(t) + \frac{\vec{a}(t) + \vec{a}(t + \Delta t)}{2}\Delta t.\end{aligned}\tag{5}$$

In this context, BOMD has advantages over the *ab initio* frequency calculation discussed above. The Stark effect is most readily observed in vibrational spectra, so using purely *ab initio* techniques we require either the second or third derivative of the electronic PES, which can be computationally expensive for larger systems. In contrast, BOMD only requires the gradient of the electronic PES to propagate nuclear position (Equations 5) and only the electron density to compute the time-dependent dipole moment (Equation 6). Thus, BOMD manages to capture some anharmonicity and higher order terms of the potential $V(\vec{r}_k)$, (Equations 1 and 2) with only the gradient for each time step.

$$\vec{\mu}(t) = - \sum_{mu} \sum_{nu} P_{\mu\nu}(\nu|\mathbf{r}(t)|\mu) + \sum_A Z_A \mathbf{R}_A(t)\tag{6}$$

Obtaining Spectra from Dipole Orientation Time Series

Since BOMD only calculates the first derivative of the PES, another method is needed to obtain spectra. The dipole vector $\vec{\mu}(t)$ is output from BOMD as a time series, so taking the first time derivative $\dot{\vec{\mu}}(t)$ and using the *dipole autocorrelation function* $\langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t + \tau) \rangle_\tau$ as the kernel of an integral transform (Equation 7) we obtain the normal mode frequencies and intensities.

$$\mathcal{A}(\omega) \propto \int \langle \dot{\vec{\mu}}(t), \dot{\vec{\mu}}(t + \tau) \rangle_\tau e^{-i\omega t} dt.\tag{7}$$

Plotting the function $\mathcal{A}(\omega)$ along the frequency domain yields a model spectrum. (696 words)

Methodology

All *ab initio* calculations performed in this work used the B3LYP/6-311G(d,p) level of theory. This density functional and basis set combination was chosen to account for polarization by including extra basis sets while still scaling efficiently with larger numbers of atoms. The static field was chosen rather than an oscillating field to simulate the local environment around each DTDCPB dimer immediately after another nearby dimer was stimulated into a charge transfer state.

First, the standard frequency calculations for the ground state of DTDCPB both with and without an applied static field were performed to determine the equilibrium geometries in the various environments and generate IR spectra in GaussView. The Stark effect was modeled by applying a static electric field of a specific magnitude generated by a dipole oriented along the z -axis normal to the plane in which the majority of the DTDCPB molecule was embedded. The spectra for the equilibrium geometries of the ground state in various field strengths were compared and tabulated to verify that the Stark effect was captured by the model and could be “calibrated” for field strength.

BOMD was first applied to a cyanide ion as a test case to coarsely model the nitriles on DTDCPB. Subsequently, it was applied to DTDCPB monomer. The BOMD calculations were carried out in zero field and also in an applied field. The systems were brought to ideal initial conditions for the “Production MD” steps by performing thermal Monte Carlo sampling subject to Boltzmann statistics in the same electric field to generate multiple trajectories for nuclear dynamics with already thermally populated vibrational normal modes. Finally, the production MD runs yielded the dipole time series for calculating the spectra. (286 words)

Computational Results

Discussion

Conclusion

Continuing work here.

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